

Credit Risk & Portfolio Optimization Dashboard

A real trained credit-risk classifier and genuine mean-variance portfolio optimization, packaged as a live dashboard — built as a runnable rebuild of a SAS-based design study, using open-source Python tooling in place of the original SAS toolset.

0.7975

TEST AUC (SELECTED MODEL)

2

CANDIDATE MODELS COMPARED

Basel III/III

REGULATORY CAPITAL CALC

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Repository	github.com/AkshitRampershad/SAS-Financial-Project
Live Dashboard	sas-financial-project-zdscpk2fdck2fggdffqhpc.streamlit.app
Stack	Python · scikit-learn · SciPy · Streamlit

Data disclosure: this system runs entirely on synthetic loan-level and market data, generated by seeded scripts in the repository (`data/generate_synthetic_loans.py` , `data/generate_synthetic_market_data.py`). No real institutional, bureau, or customer data is used anywhere in this project. Every metric in this report is computed live from that synthetic data — none are hardcoded or tuned to hit a target number.

1. Executive Summary

This project rebuilds a credit-risk and portfolio-optimization design study — originally scoped around SAS/STAT, SAS/OR, and SAS/ETS — as a real, runnable Python system. Rather than simulate the SAS toolset's outputs, every step in the original methodology is re-implemented with an open-source equivalent and measured on synthetic data generated specifically for this project:

- A genuine credit-risk classifier is trained and evaluated on held-out data, with a second candidate model trained alongside it purely for comparison and model selection — not for show.
- Real mean-variance portfolio optimization (Markowitz) computes the efficient frontier, the maximum-Sharpe-ratio portfolio, and a risk-tolerance-driven allocation, using a numerical solver rather than a lookup table.
- A simplified but standard Basel II/III Internal Ratings-Based (IRB) capital formula converts the credit book's risk into a regulatory capital requirement.
- Three CCAR-style recession scenarios (mild, moderate, severe) are used to re-score the credit book and shock the investment portfolio, and report the resulting change in both.

The result is packaged as a four-tab Streamlit dashboard, deployed live, that a reviewer can interact with directly rather than take on faith from a static writeup.

2. From SAS to Python: What Changed

The original design study specified a SAS/STAT + SAS/OR + SAS/ETS toolchain. Every one of those components has a direct, real (not simulated) equivalent in this rebuild:

Original SAS Component	This Project's Implementation
PROC LOGISTIC , PROC DISCRIM	scikit-learn <code>LogisticRegression</code> , compared against a <code>GradientBoostingClassifier</code> and selected by actual held-out test AUC.
PROC HPFOREST , SAS Enterprise Miner	The gradient-boosting candidate stands in for the ensemble method; selection is logged as a small local model-comparison run.
PROC OPTMODEL	Real mean-variance optimization via <code>scipy.optimize</code> (SLSQP) — efficient frontier, max-Sharpe tangency portfolio, capital-allocation-line risk-tolerance sweep.
SAS/ETS, PROC ARIMA	A synthetic but structured macro cycle (unemployment, GDP growth, with a modeled 2020-style shock) feeds the credit model's origination-time features and the stress scenarios.
Regulatory capital calculation	A real, simplified Basel II/III IRB capital formula — not a placeholder number.
SAS Visual Analytics	This Streamlit dashboard.
2.5M real loan records, bureau/Bloomberg/FRED data	A synthetic, seeded ~30,000-loan portfolio and synthetic multi-asset return series — not scraped or real institutional data.

3. Data

3.1 Synthetic Loan Portfolio

A seeded generator produces a portfolio of loans with realistic origination-time features, used to train and evaluate the credit-risk model:

Feature group	Fields
Loan terms	loan amount, term (months), interest rate, secured flag, loan purpose
Borrower profile	credit score, annual income, debt-to-income ratio, employment length, delinquencies (2yr), home ownership, industry sector, region
Macro context at origination	unemployment rate, GDP growth rate
Target	default flag (binary)

3.2 Synthetic Market Data

A separate generator produces monthly returns for a ten-asset-class universe (US large-cap, small-cap, international developed, and emerging-market equities; investment-grade and high-yield bonds; Treasuries; REITs; commodities; and cash), each driven by factor loadings on equity, rate, and inflation shocks plus idiosyncratic monthly volatility — a structured simulation, not random noise.

4. Credit Risk Model

Two candidate classifiers are trained on an identical preprocessing pipeline (median imputation and standard scaling for numeric features, one-hot encoding for categorical features) and evaluated on a held-out 25% test split:

Candidate	Test AUC	Test Accuracy	Mean Predicted PD
Logistic Regression (<i>selected</i>)	0.7975	measured, not tuned	≈ matches true base rate
Gradient Boosting	0.7965	measured, not tuned	—

The two models scored within 0.001 AUC of each other; logistic regression was selected on that measured result, not preferred in advance.

A CALIBRATION DECISION, NOT JUST AN ACCURACY ONE

`class_weight="balanced"` was deliberately **not** used. It reweights training toward a roughly 50/50 class prior, which distorts `predict_proba` into a no-longer-genuine probability — verified directly: it inflated the model's mean predicted default probability from about 4.9% (matching the true base rate) to roughly 35%. Since this model's entire purpose is a calibrated probability of default for `Expected Loss = PD × LGD × EAD`, a well-ranked but miscalibrated score isn't good enough — and the unweighted model's AUC was, if anything, marginally better anyway.

4.1 Feature Importance

Feature importance is extracted directly from the selected model — tree-based `feature_importances_` for gradient boosting, or the absolute logistic-regression coefficients (normalized) as an influence proxy — and surfaced in the dashboard's Credit Risk tab, alongside a per-loan-purpose and per-credit-score-band risk breakdown, an ROC curve, and an interactive "score a hypothetical loan" tool.

5. Portfolio Optimization

Mean-variance (Markowitz) optimization is computed with `scipy.optimize.minimize` using the SLSQP solver, subject to a fully-invested, long-only constraint and a 40% maximum weight per asset — the same kind of constraint set a real institutional mandate would impose.

- **Efficient frontier:** 25 minimum-variance portfolios computed across the achievable return range.
- **Maximum-Sharpe (tangency) portfolio:** the risky-asset allocation that maximizes the Sharpe ratio against a 2.5% risk-free rate.
- **Risk-tolerance slider:** a two-fund (capital allocation line) split between cash and the tangency portfolio, following Tobin separation.

WHY TWO-FUND SEPARATION, NOT ONE JOINT OPTIMIZATION

A single mean-variance optimization over all assets *including* cash was deliberately avoided: a near-zero-volatility asset's Sharpe ratio is dominated by its tiny denominator, so a joint optimizer tends to load up on cash regardless of the stated risk tolerance — a well-known pitfall, not a useful "risk tolerance" story. Separating the risk-free asset out and varying only the cash/risky-portfolio split by risk aversion is both the textbook-correct approach and what actually produces a sensible aggressive-to-conservative slider.

Expected returns feeding the optimizer are analytic assumptions computed from the market-data generator's own factor parameters — not a noisy trailing sample mean. With realistic volatility and only a decade of simulated monthly history, a trailing mean's standard error would swamp the signal, which is exactly why real institutions use separate capital-market assumptions rather than historical averages for return forecasts.

6. Regulatory Capital & Stress Testing

6.1 Basel II/III IRB Capital

Per-unit-exposure unexpected-loss capital is computed from the real Basel Internal Ratings-Based formula:

$$K = LGD \times [N((N^{-1}(PD) + \sqrt{R} \times N^{-1}(0.999)) / \sqrt{(1-R)}) - PD]$$

where N is the standard normal CDF. Required capital is this per-unit factor times each loan's exposure at default, scaled to risk-weighted assets against the Basel III 8% minimum capital ratio. The one deliberate simplification: Basel's real retail/corporate formula uses a PD-dependent asset correlation curve (roughly 0.03–0.24); this uses a single fixed correlation of 0.12, near the middle of that range, documented directly in the code rather than left implicit.

6.2 Stress Scenarios

Three CCAR-style severity tiers re-score the existing loan book's probability of default through the real trained model with shocked macro inputs, and apply a one-time return shock per asset class to the investment portfolio:

Scenario	Unemployment shock	GDP growth shock	US Large-Cap Equity shock
Mild Recession	+1.5 pp	−2.0 pp	−10%
Moderate Recession	+3.0 pp	−4.0 pp	−20%
Severe Recession	+5.0 pp	−6.5 pp	−35%

Both the credit and portfolio stress paths are genuine re-evaluations against the scenario's own shocked numbers — not a canned "apply -20% to everything" shortcut.

7. Honesty Principles

Two disclosure principles are held throughout, and stated explicitly in the dashboard itself:

1. The credit model's AUC and calibration are whatever they actually measure on held-out data — never tuned to hit a target number.
2. The portfolio optimizer's expected returns are analytic assumptions from the generator's own parameters, not a noisy trailing sample mean, for the statistical reason given in Section 5.

8. Tech Stack & Repository

Language / Runtime	Python 3.11+
ML & Optimization	scikit-learn, SciPy (optimize, stats)
Data & App	pandas, NumPy, Streamlit, Plotly
Testing	pytest suite covering the data generators, the credit model, the optimizer, and stress-test invariants
Repository	github.com/AkshitRampershad/SAS-Financial-Project
Live Dashboard	sas-financial-project-zdscpk2fdck2fggdffqhpc.streamlit.app

This report describes a personal portfolio project built on synthetic data for demonstration purposes. It is not the output of, and does not represent, any real financial institution's credit or investment process.